

Tensor computations

Near-optimal type-2 bound for Gaussian sums of symmetric tensors

Difficulty: extreme **Importance:** broadly interesting**Status:** Partially resolved

Rating rationale: Extreme because the low- p regime encounters an explicit geometric barrier in tensor concentration; broad importance spans probability, analysis, theoretical computer science and noisy tensor computation.

Statement

For a symmetric real tensor $T \in (\mathbb{R}^d)^{\otimes r}$, define

$$\|T\|_{J_p} = \max_{\|x\|_p \leq 1} |\langle T, x^{\otimes r} \rangle|.$$

For every integer $r \geq 2$ and real $2 \leq p < \infty$, do constants $C_{r,p} > 0$ and $a_{r,p} \geq 0$ exist such that, for every $d \geq 2, N \geq 1$, and deterministic symmetric tensors $T_1, \dots, T_N \in (\mathbb{R}^d)^{\otimes r}$,

$$\mathbb{E} \left\| \sum_{i=1}^N g_i T_i \right\|_{J_p} \leq C_{r,p} d^{1/2-1/p} [\log(2 + dN)]^{a_{r,p}} \left(\sum_{i=1}^N \|T_i\|_{J_p}^2 \right)^{1/2},$$

where g_i are independent real $N(0, 1)$ random variables? Constants and logarithmic exponents must be independent of d, N and the tensors. The explicit logarithmic form spells out the source's $\tilde{O}_{r,p}$ notation.

Relevance

This would control the injective norm of tensor perturbations with general covariance, extending a central matrix concentration principle to multilinear computation and noisy tensor models.

References

1. A. S. Bandeira, D. Dmitriev, K. Lucca, P. Nizić-Nikolac, and A. Rödder, *Randomstrasse101: Open Problems of 2025*, arXiv:2603.29571, 2026. [Primary text](#), Entry 8 by K. Lucca, Conjecture 16 and equations (1)–(2).
2. A. S. Bandeira, S. Gopi, H. Jiang, K. Lucca, and T. Rothvoss, *A Geometric Perspective on the Injective Norm of Sums of Random Tensors*. [Primary preprint](#), introduction and tensor type-2 bounds; published as *Tensor Concentration Inequalities: A Geometric Approach*, STOC 2025.

Status check — 2026-09-10

Rechecked [Randomstrasse101](#), [Entry 8](#), [Conjecture 16](#) and the [underlying tensor-concentration paper](#), then searched for later type-2 bounds. The displayed inequality is proved for p at least twice the tensor order and for the matrix case $r=p=2$; the general low- p regime remains unresolved in the August 2026 manuscript. No full resolution was located. Restrictions on covariance or summands do not establish the universal claim.